Atropine

- Atropine is an anticholinergic which is mostly used for the treatment of bradycardia.
- Atropine antagonises (blocks) muscarinic acetylcholine receptors, causing vagal inhibition resulting in:
  - An increase in heart rate.
  - Drying of salivary and bronchial secretions.
  - Bronchodilation.
  - Reduced gastrointestinal motility.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Adults with bradycardia, particularly if the rhythm is narrow complex.
- ✓ Organophosphate poisoning.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Myocardial ischaemia. Atropine will increase myocardial oxygen consumption.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 0.6 mg for an adult. Repeat as required without a maximum dose, if the bradycardia is responsive to atropine.

Atropine

- Repeated and escalating doses are likely to be required for organophosphate poisoning.

Administration

- Administer undiluted as a rapid IV bolus. Slow administration may result in transient bradycardia.

Common adverse effects

- Tachycardia.
- Confusion. Particularly in the elderly or those with intellectual impairment.
- Dry mouth.
- Blurred vision.

Usual onset of effect

- 5-10 seconds.

Usual duration of effect

- The cardiovascular effects last 15-60 minutes.
- The exocrine and smooth muscle effects last 4-6 hours.

Usual preparation

- Ampoule containing 0.6 mg in 1 ml.

Pharmacokinetics

- Atropine is predominantly metabolised in the liver, but some is excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

Atropine

- The action of atropine may be potentiated if the patient is taking other drugs with anticholinergic properties, such as phenothiazines, some antihistamines (such as promethazine, but not loratadine), tricyclic antidepressants and anti-parkinsonian medicines. These interactions are rarely clinically significant.

More Information

Version history

11 September 2023 1.0.4

Update: Indications updated to align with changes to bradycardia guideline.